

2025-2026 Academic Year
Non parametric Statistics course project

Identifying Alzheimer's Disease through brain volumes longitudinal data

A study about Alzheimer Neurodegenerative Dementia influence on human brain volumes

Martina Caliandro 10797483
Greta Minazzi 10765748
Matteo Zanetti 10765931

Tutor:
Dr. Soumick Chatterjee

Research Question

Is it possible to develop a more robust method to identify high-risk patients for Alzheimer's disease using longitudinal brain volumes?

Why?

Objective
diagnosis



Robust
monitoring



Better care
planning



Dataset

- 1176 patients - longitudinal structure
- 117 brain regions volumes extracted from MRI images
- 3 conditions of Alzheimer Dementia:



Cognitive Normal (CN)

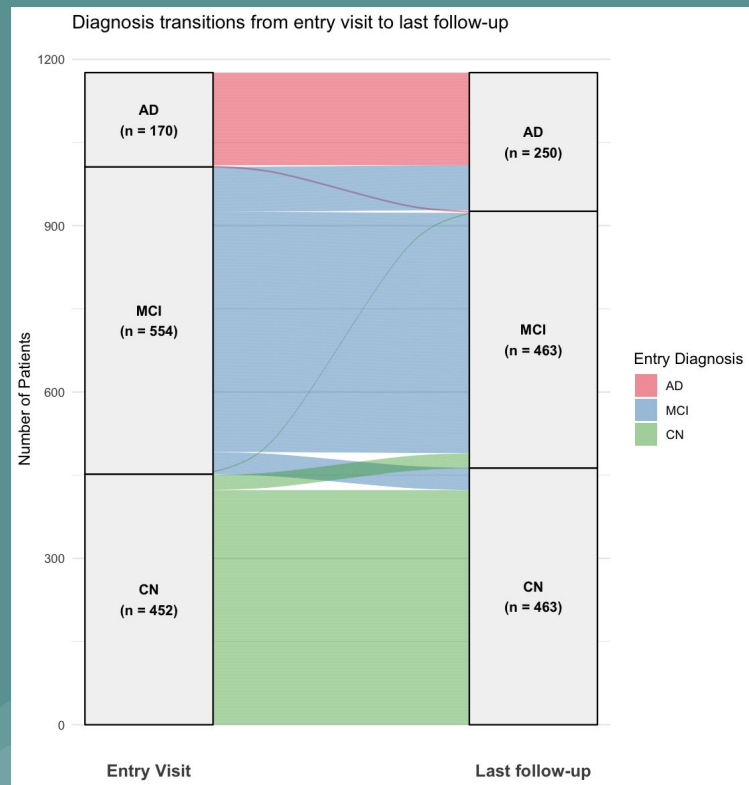


Mildly Cognitive Impaired (MCI)

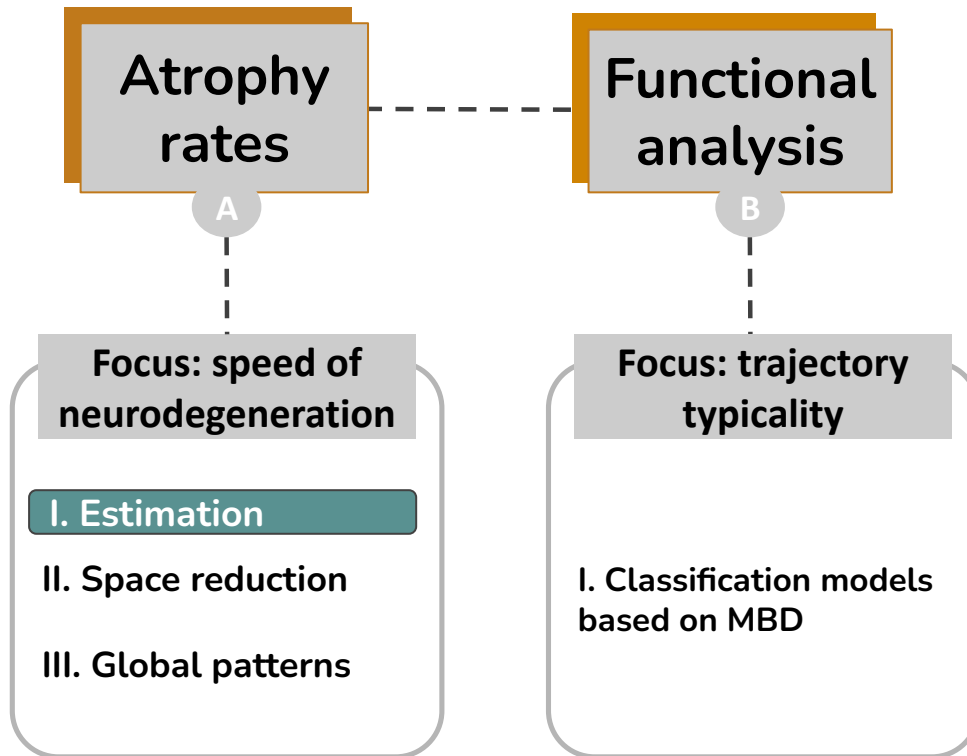


Alzheimer Disease (AD)

- Additional clinical features (e.g. age, genotype)



Methodological roadmap



Quantile Regression

$$y_{ij} = \beta_{0,i} + \beta_{1,i}t_{ij} + \epsilon_{ij} \quad i = 1, \dots, I \quad j = 1, \dots, n_i$$

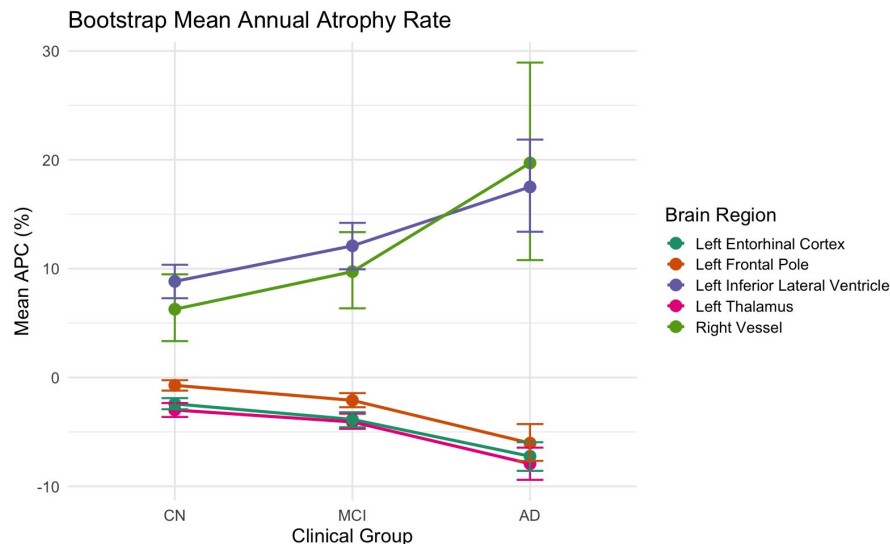
$y_{i,j}$: volume of patient i at visit j , $t_{i,j}$: years from baseline of visit j for patient i

I : number of patients, n_i : number of visits for patient i

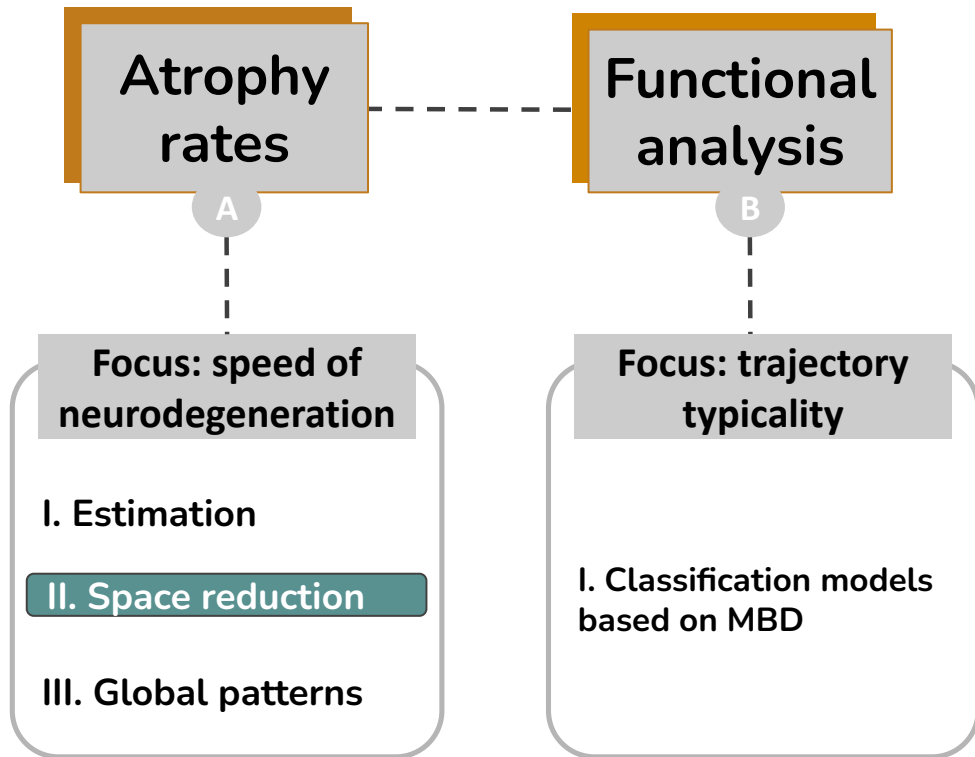
$$\text{Atrophy Rate}_{\%}^{(i)} = \frac{\hat{\beta}_{1,i}}{\hat{\beta}_{0,i}} \times 100$$

Quantile Regression: validation of atrophy rates

- ❑ **Global univariate permutational independent tests** for each atrophy rate, using Benjamin-Hochberg correction
- ❑ **Bootstrap 95% CI** for the mean atrophy rate (volume specific) in each diagnostic group
- ❑ **Key findings:** estimated atrophy rates align with literature-reported vulnerable brain regions in AD

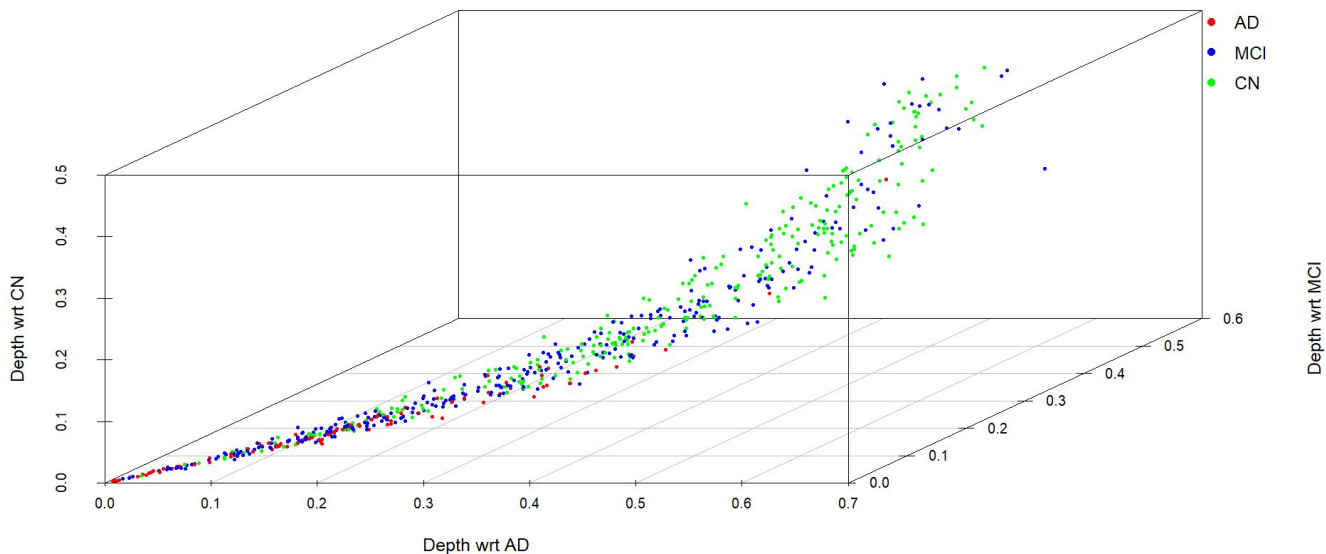


Methodological roadmap



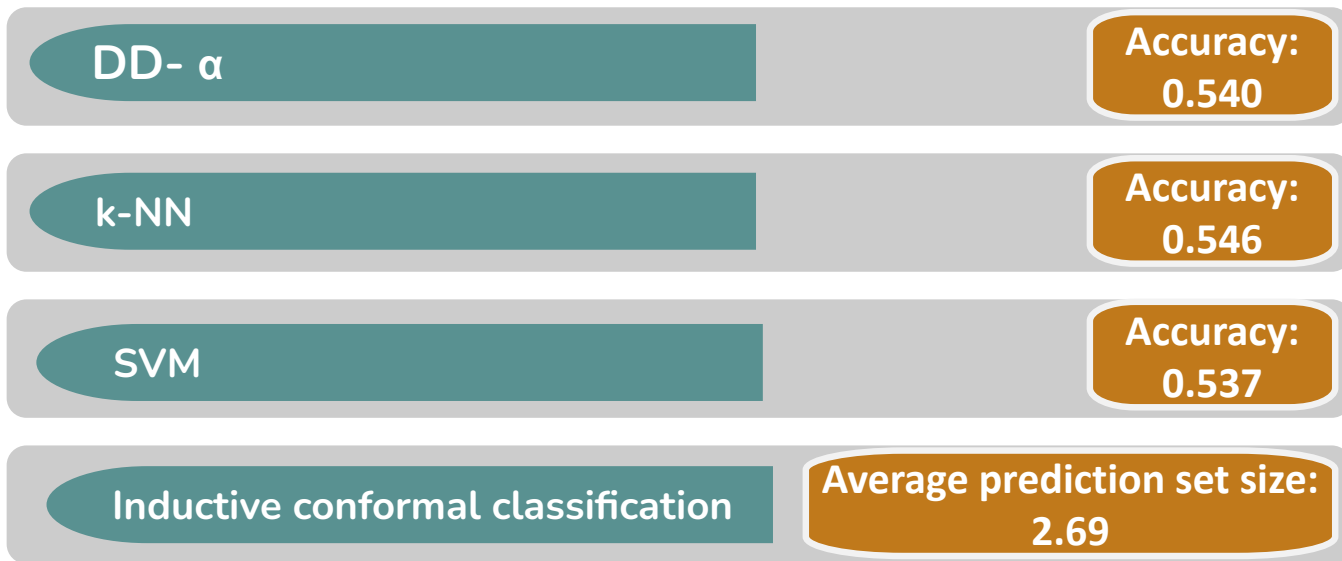
Depth Space:

$$f : \mathbb{R}^p \rightarrow \mathbb{R}^3, \quad f(X_i) = (D_{CN}(X_i), D_{MCI}(X_i), D_{AD}(X_i))$$



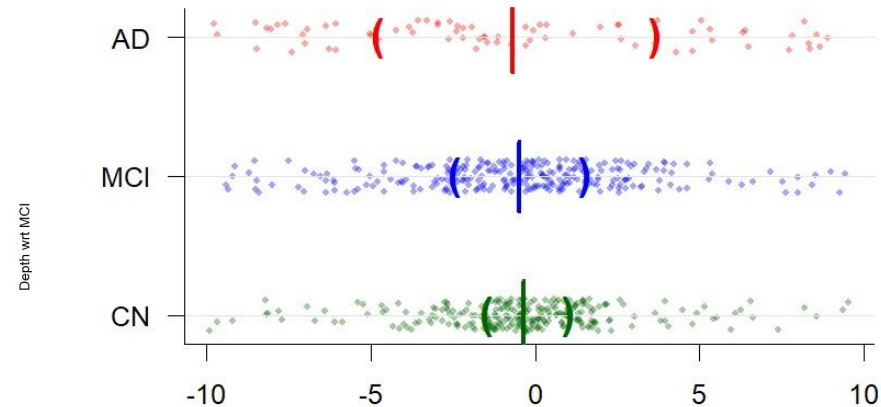
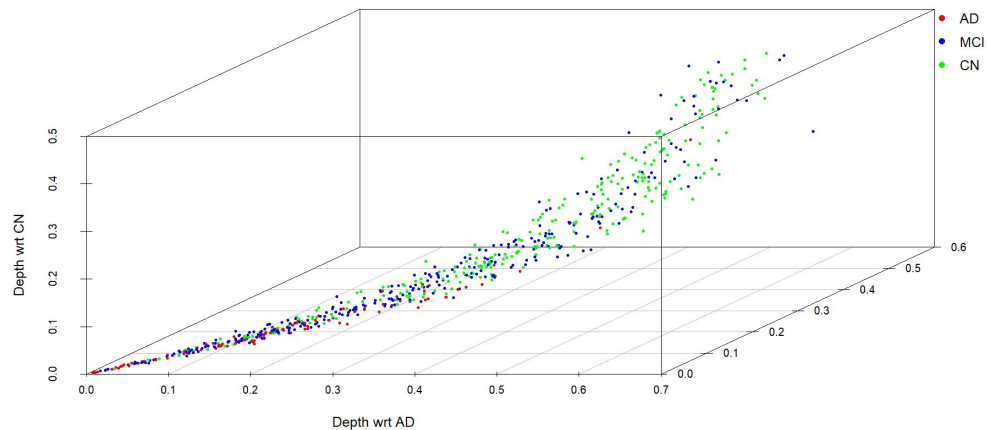


Depth Based Classification

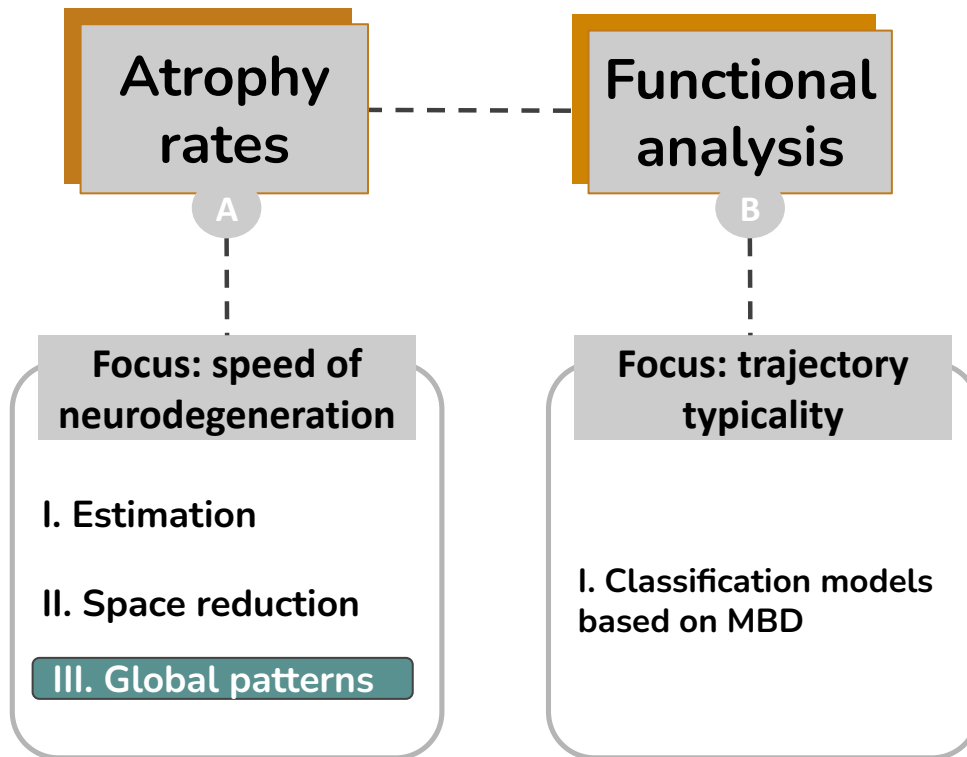




Closer inspection of the data



Methodological roadmap



Dispersion quantifiers

Mean absolute deviation

$$T_i^{(1)} = \frac{1}{p} \sum_{j=1}^p |Z_{ij}|$$

CV accuracy:
0.557

Variance of absolute deviations

$$T_i^{(2)} = \text{Var}(|Z_{ij}|)$$

CV accuracy:
0.539

Skewness of Absolute Deviations

$$T_i^{(3)} = \text{Skewness}(|Z_{ij}|)$$

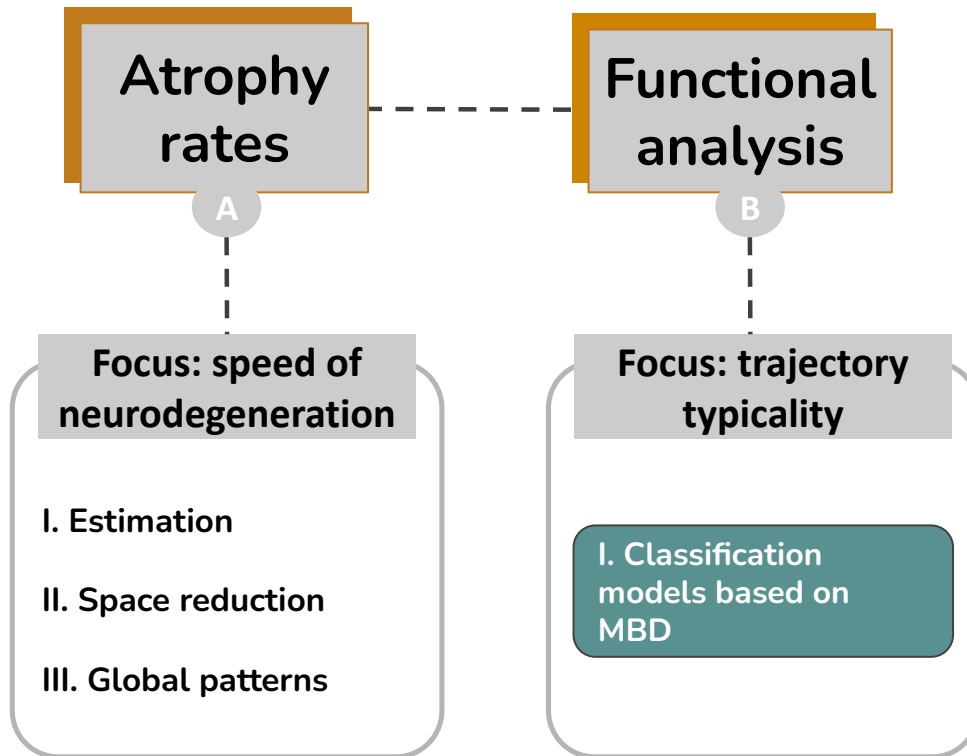
CV accuracy:
0.432

Correlation-Pattern Deviation

$$T_i^{(4)} = Z_i^\top \Sigma_{CN}^{-1} Z_i$$

CV accuracy:
0.600

Methodological roadmap





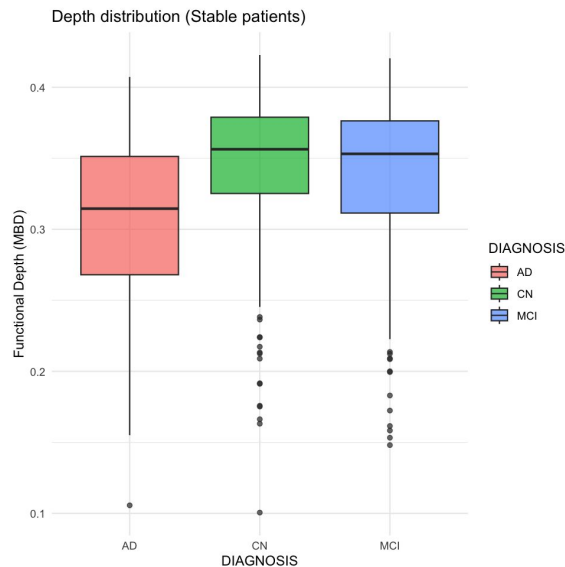
Functional depth-based classification

$$\text{Functional Depth}_i = \frac{1}{V} \sum_{v=1}^V \text{MBD}\left(f_i^{(v)}(t_1), \dots, f_i^{(v)}(t_L)\right)$$

$i = 1, \dots, N$ (patients) $v = 1, \dots, V$ (volumes)

$t_1, \dots, t_L$ (age grid) $f_i^{(v)}(t)$ = interpolated trajectory

MBD: modified band depth



Functional depth-based classification

- I. Max depth classifier: $\hat{y}_i = \arg \max_{k \in \mathcal{G}} \{D_{\text{CN}}(X_i), D_{\text{MCI}}(X_i), D_{\text{AD}}(X_i)\}$

3-class model
(full brain - 117
regions)

CV accuracy (stable): 0.449
Accuracy (converters): 0.419

3-class model
(Selected Markers - 18
Regions)

CV accuracy (stable): 0.472
Accuracy (converters): 0.362



Functional depth-based classification

II. **Logistic regression:** $\text{IP}(\text{AD}) = \text{logit}^{-1} (\beta_0 + \beta_1 \cdot \overline{\text{depth}}_i + \beta_2 \cdot \text{age}_i + \beta_3 \cdot \text{GENOTYPE}_i)$

$$\overline{\text{depth}}_i = \frac{1}{p} \sum_{j=1}^p \text{MBD}(X_{ij} \mid G_{CN,j}) \quad i = 1, \dots, I \quad j = 1, \dots, p$$

X_{ij} : functional trajectory of region j for subject i

$G_{CN,j}$: reference set of healthy trajectories for region j

I : total number of subjects, p : total number of brain regions



Main Achievements: Evaluating the Logistic Model

$$\text{IP(AD)} = \text{logit}^{-1} (\beta_0 + \beta_1 \cdot \overline{\text{depth}_i} + \beta_2 \cdot \text{age}_i + \beta_3 \cdot \text{GENOTYPE}_i)$$

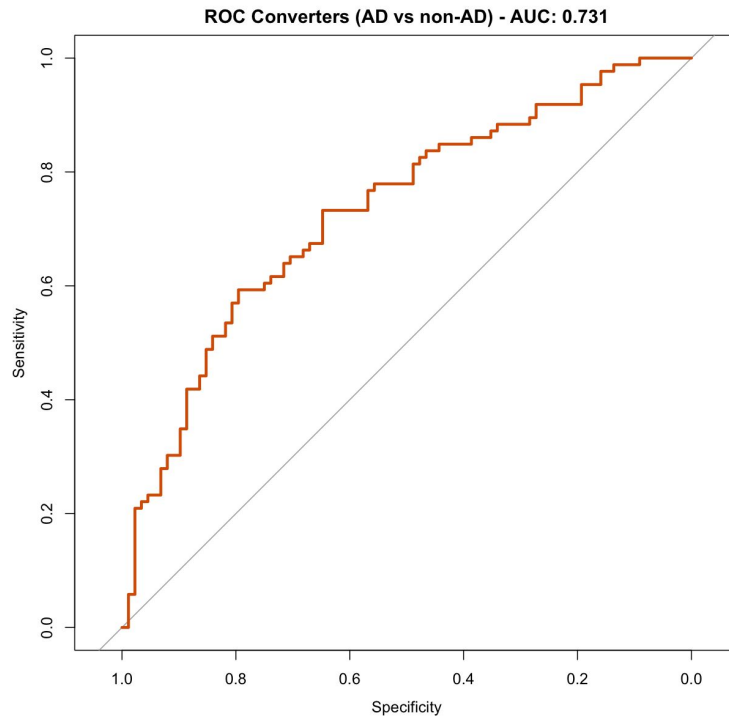
Predictor	β	Std.Error	zvalue	Pvalue
(Intercept)	-3.958	1.424	-2.780	0.0054
functional_depth	-10.004	1.898	-5.271	$1.35 \cdot 10^{-7}$
age	0.062	0.017	3.643	0.0003
GENOTYPE 2/2	14.632	535.411	0.027	0.9782
GENOTYPE 2/3	0.396	0.522	0.759	0.4479
GENOTYPE 2/4	2.041	0.729	2.799	0.0051
GENOTYPE 3/4	1.371	0.279	4.924	$8.50 \cdot 10^{-7}$
GENOTYPE 4/4	2.322	0.367	6.331	$2.43 \cdot 10^{-10}$

Main Achievements: Evaluating the Logistic Model

CV accuracy (stable): 0.8476

Accuracy (converters): 0.592

Optimized threshold
Accuracy (converters): 0.695



Possible further steps

Neural Networks

Involve a subject expert (a neurologist)

Larger cohort for more stable estimates

**More converters-patients to evaluate the
diagnose**



Bibliography

- **Pokotylo, O., Mozharovskyi, P., & Dyckerhoff, R. (2019)**, Depth and depth-based classification with R package ddalpha. *Journal of Statistical Software*, 91(5), 1–46.
- **Lange, T., Mosler, K., & Mozharovskyi, P. (2014)**, Fast nonparametric classification based on data depth. *Statistical Papers*, 55, 49–69.
- **López-Pintado, S., & Romo, J. (2009)**, On the concept of depth for functional data. *Journal of the American Statistical Association*, 104(486), 718-734.
- **Sun, Y., & Genton, M. G. (2011)**, Functional boxplots. *Journal of Computational and Graphical Statistics*, 20(2), 316-334.
- **Fluss, R., Faraggi, D., & Reiser, B. (2005)**, Estimation of the Youden Index and its associated cutoff point. *Biometrical Journal*, 47(4), 458-472.
- **Zuo, Y. (2003)**, Projection-based depth through multivariate location and scale. *The Annals of Statistics*, 31(5), 1460-1490.

**Thank you for your
attention!**

The background is a solid teal color. It features several faint, semi-transparent geometric shapes. In the upper right, there is a large circle containing a pie chart with three segments. To its right and below are several smaller circles, each also containing a pie chart. In the bottom right corner, there is a bar chart with four vertical bars of increasing height from left to right. The text "Thank you for your attention!" is centered in the middle of the image in a bold, white, sans-serif font.